



THE UNIVERSITY OF
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ADSP 31014

Statistical Models for Data Science

Multiple Linear Regression

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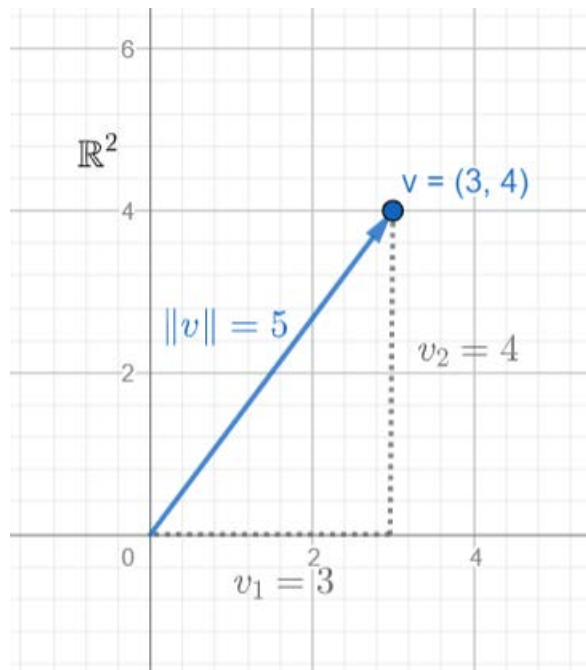
L2-Norm

- Norm = length of vector = distance from origin
- **Pythagorean theorem:**

$$\vec{v} = (v_1, \dots, v_d) \in \mathbb{R}^d$$

$$\|\vec{v}\| = \sqrt{(v_1)^2 + \dots + (v_d)^2}$$

- In 1-dim real number line $\mathbb{R}$, norm is just the absolute value $|v|$



Simple Linear Regression

- Observed data: $(x_1, y_1), \dots, (x_n, y_n)$
- Model equation:

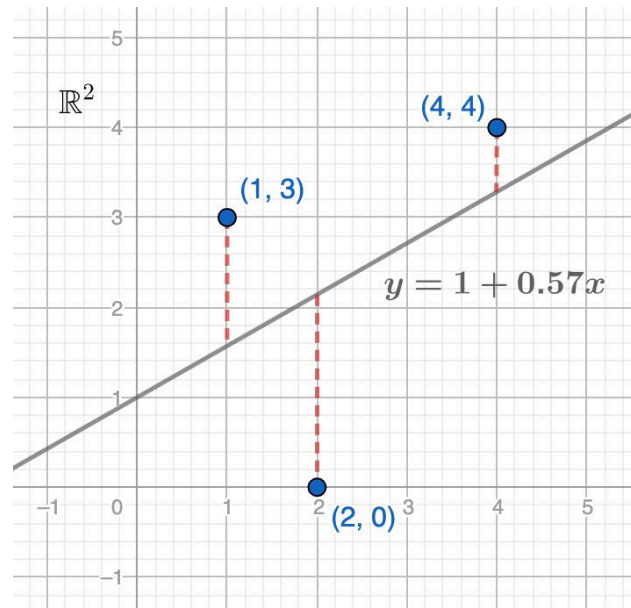
$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad \text{for } i = 1, \dots, n$$

line

intercept slope noise term

- Find optimal β_0^* and β_1^* such that the line fits the data best

$$y_i \approx \beta_0^* + \beta_1^* x_i \quad \text{for } i = 1, \dots, n$$



- Least squares minimization:** Find optimal β_0^* and β_1^* to

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n \overbrace{|y_i - (\beta_0 + \beta_1 x_i)|^2}^{\text{vertical discrepancy}}$$

Vector Formulation

- Write the response, intercept, and explanatory data as vectors in $\mathbb{R}^n$

$$\vec{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \quad \vec{\mathbf{1}} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \quad \vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

$$\begin{aligned} & \sum_{i=1}^n |y_i - (\beta_0 + \beta_1 x_i)|^2 \\ &= \left\| \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} - \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_0 \end{pmatrix} - \beta_1 \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right\|^2 \\ &= \|\vec{y} - (\beta_0 \vec{\mathbf{1}} + \beta_1 \vec{x})\|^2 \end{aligned}$$

- Find optimal β_0^* and β_1^* to

$$\min_{\beta_0, \beta_1} \|\vec{y} - (\beta_0 \vec{\mathbf{1}} + \beta_1 \vec{x})\|$$

- Recall L_2 -norm $\|\vec{v}\|^2 = \sum_{i=1}^d |v_i|^2$

Multiple Linear Regression

- n data points but $p > 1$ explanatory variables
- Each i^{th} observed data:

$$y_i \in \mathbb{R} \quad \vec{x}_i = (x_{i1}, \dots, x_{ip}) \in \mathbb{R}^p$$

- Model equation:

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i$$

for $i = 1, \dots, n$

- Rewrite the minimization in matrix form

$$\min_{\beta_0, \dots, \beta_p} \sum_{i=1}^n |y_i - (\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip})|^2 = \min_{\vec{\beta}} \|\vec{y} - X\vec{\beta}\|^2$$

Matrix Form

- Rewrite together as

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \approx \beta_0 + \beta_1 \begin{pmatrix} x_{11} \\ \vdots \\ x_{n1} \end{pmatrix} + \dots + \beta_p \begin{pmatrix} x_{1p} \\ \vdots \\ x_{np} \end{pmatrix}$$
$$= \underbrace{\begin{pmatrix} | & x_{11} & \dots & x_{1p} \\ \vec{1} & \vdots & & \vdots \\ | & x_{n1} & \dots & x_{np} \end{pmatrix}}_{\text{design matrix } X \in \mathbb{R}^{n \times (1+p)}} \times \underbrace{\begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}}_{\vec{\beta} \in \mathbb{R}^{1+p}}$$

Matrix Calculus

- **Question:** How to solve the least squares problem for $\vec{\beta}^*$ algebraically?

$$\min_{\vec{\beta}} \|\vec{y} - X\vec{\beta}\|^2$$

- **Step 1:** Rewrite as

$$\begin{aligned} f(\vec{\beta}) &= \|\vec{y} - X\vec{\beta}\|^2 \\ &= (\vec{y} - X\vec{\beta})^T (\vec{y} - X\vec{\beta}) \\ &= (\vec{y}^T - \vec{\beta}^T X^T)(\vec{y} - X\vec{\beta}) \\ &= \underbrace{\vec{\beta}^T X^T X \vec{\beta} - 2\vec{\beta}^T X^T \vec{y} + \vec{y}^T \vec{y}}_{\text{quadratic function in variable } \vec{\beta}} \end{aligned}$$

- **Step 2:** To minimize $f(\vec{\beta})$, use the first-order condition

$$\frac{\partial}{\partial \vec{\beta}} f(\vec{\beta}) = \vec{0}$$

In 1-dimensional case,

$$\frac{d}{dx} (ax^2 + bx + c) = 2ax + b$$

In high-dimensional case, use matrix calculus

$$\begin{aligned} \frac{\partial}{\partial \vec{\beta}} (\vec{\beta}^T X^T X \vec{\beta} - 2\vec{\beta}^T X^T \vec{y} + \vec{y}^T \vec{y}) \\ = 2X^T X \vec{\beta} - 2X^T \vec{y} \end{aligned}$$

Normal Equation

- **Step 3:** A solution $\vec{\beta}^*$ of

$$\min_{\vec{\beta}} \|\vec{y} - X\vec{\beta}\|^2$$

is a solution of the **normal equation**

$$X^T X \vec{\beta} = X^T \vec{y}$$

Diagram illustrating the dimensions of the matrices and vectors in the normal equation:

- X^T is a $(1+p) \times n$ matrix.
- X is an $n \times (1+p)$ matrix.
- $\vec{\beta}$ is a $(1+p) \times 1$ vector.
- $X^T \vec{y}$ is an $n \times 1$ vector.
- $\vec{y}$ is a $1 \times n$ vector.
- The equation is written as $\underbrace{X^T X}_{A \in \mathbb{R}^{(1+p) \times (1+p)}} \vec{\beta} = \underbrace{X^T \vec{y}}_{\vec{b} \in \mathbb{R}^{1+p}}$.

- This is a system of $1 + p$ linear equations in $1 + p$ unknown variables $\vec{\beta}$

Formula for $\vec{\beta}^*$

- If $X^T X \in \mathbb{R}^{(1+p) \times (1+p)}$ is invertible, then the linear regression has unique **fitted model parameters**

$$\vec{\beta}^* = (X^T X)^{-1} X^T \vec{y}$$

- Fitted values / predictions

$$\begin{aligned} \vec{y}^* &= X \vec{\beta}^* \\ &= X (X^T X)^{-1} X^T \vec{y} \end{aligned}$$

- **Question:** But when is $X^T X$ invertible?